GTIIT Spring 2025 Calculus 1M

Workshop 2a

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Lecture 3

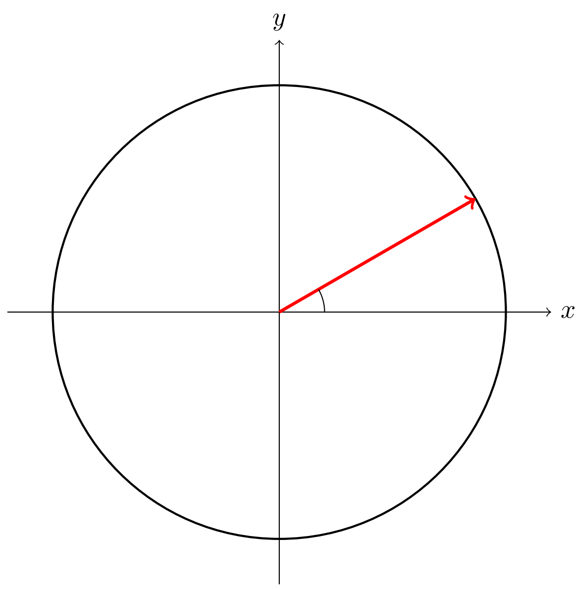
1. (i) Sketch the graph of the circle with equation $x^2 + y^2 + x = 0$.

   (ii) Under what conditions does the equation $x^2 + y^2 + ax + by + c = 0$ represent a circle?

   (iii) Identify the type of curve represented by the equation $x^2 + y^2 + x = 0$ and sketch its graph.

   (iv) Sketch the region bounded by the curves $y = 4 - x^2$ and $y = x^2 - 4$.

2. Draw and label in the following circle the geometric representation of the sine, cosine and tangent functions for an angle θ in the first quadrant.



3. (i) Solve the equation

$$\sin x = \cos x$$

   for x in the interval $[0, 2\pi)$. *hint:* Divide both sides by $\cos x$ (assuming $\cos x \neq 0$).

(ii) Prove the identity

$$\tan x + \cot x = \frac{2}{\sin 2x}.$$

hint: Use the double-angle formula $\sin 2x = 2\sin x\cos x$.

(iii) Solve the equation

$$\sin 2x = \sqrt{3}\cos x$$

for x in $[0, 2\pi)$. *hint:* Use the identity $\sin 2x = 2\sin x\cos x$ and consider the cases $\cos x \neq 0$ and $\cos x = 0$.

(iv) Show that the expression $\sin x + \cos x$ can be rewritten as

$$\sin x + \cos x = \sqrt{2}\sin\left(x + \frac{\pi}{4}\right).$$

hint: Recall the sine addition formula

$$\sin\left(x + \frac{\pi}{4}\right) = \sin x\cos\frac{\pi}{4} + \cos x\sin\frac{\pi}{4}.$$

(v) Prove the double-angle formula

$$\cos 2x = 2\cos^2 x - 1.$$

hint: Use the standard double-angle formula $\cos 2x = \cos^2 x - \sin^2 x$.

Lecture 4

1. Find the domain and range of the following functions:
   (a) $f(x) = \sqrt{x^2 - 4}$
   (b) $g(x) = \frac{1}{x^2-4}$
   (c) $h(x) = \frac{1}{\sqrt{x^2-4}}$
2. Sketch the graph of the following functions:
   (a) $f(x) = \begin{cases} x^2 & \text{if } x \leq 0 \\ 2x & \text{if } x > 0 \end{cases}$
   (b) $g(x) = \begin{cases} |x| & \text{if } x \leq 1 \\ 1 & \text{if } x > 1 \end{cases}$
   (c) $h(x) = ||x| - 1|$
3. Determine whether f is even, odd, or neither:
   (a) $f(x) = x^3 - x$
   (b) $g(x) = \frac{1}{x^2+1}$
4. If f and g are both even functions, what can you say about $f + g$ and $f \cdot g$? What if f is even and g is odd?